

SECD2613

System Analysis and Design



PART V #1

Output, Input and User Interface Design

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■ OBJECTIVES

- Understand the objectives for the effective input/output design and user interface design.
- Design functional input forms for different users of business systems.
- Relate output content to output methods inside and outside the organization.
- Design useful input forms and display output for people interacting on the web (variety of user interfaces).

■ MAJOR TOPICS

OUTPUT DESIGN

- Output classification
- Reports
- Output screen design

INPUT DESIGN

- Input methods
- Design guidelines

USER INTERFACE DESIGN

- Guidelines for user interface design
- Storytelling HCI

■ INTRODUCTION

- Output and user interface design is the first task in the systems design phase of the SDLC

OUTPUT DESIGN focuses on **user needs for screen and printed forms of output**, while **USER INTERFACE DESIGN** stresses **user interaction with the computer**, including **INPUT DESIGN** and procedures

■ MAJOR TOPICS

OUTPUT DESIGN

- Output classification
- Reports
- Output screen design

■ OUTPUT DESIGN

- Before designing output, ask yourself several questions:

What is the purpose of the output?

Who wants the information, why it is it needed, and how will it be used?

What specific information will be included?

Will the output be printed, viewed on-screen, or both? What type of device will the output go to?



Note:

Your answers to these questions will affect your output design strategies

- The quality of system input will determines the quality of system output.
- Output can be in many forms such as printout, reports, display audio, CD's, e-mail or webpages.

■ OUTPUT DESIGN OBJECTIVES

- Output is essential to ensuring the use and acceptance of the information system.

1. Designing output to serve the intended purpose

2. Designing output to fit the user?

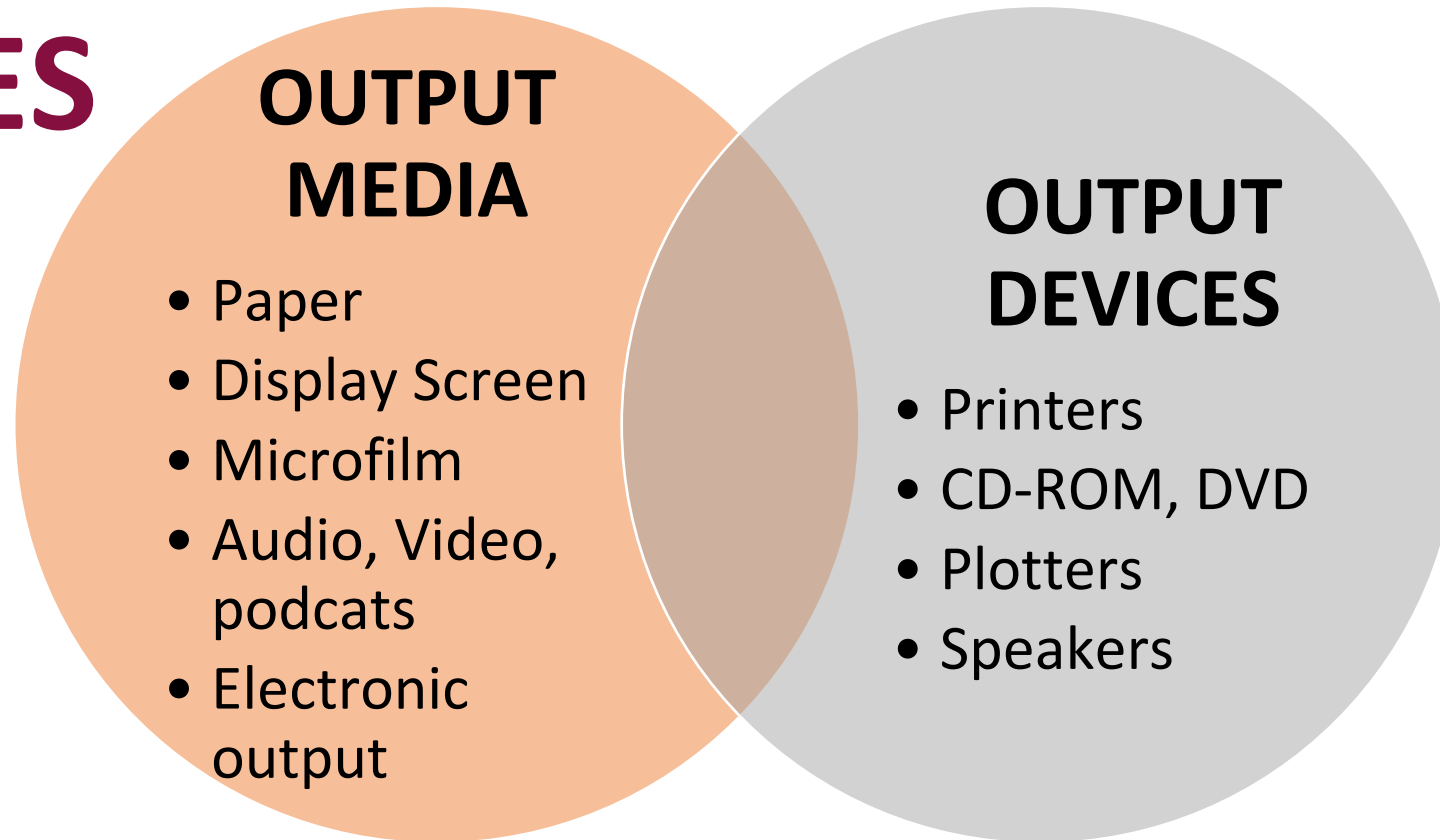
3. Delivering the appropriate quantity of output

4. Making sure the output is where it is needed

5. Providing the output on time

6. Choosing the right output method

■ OUTPUT TECHNOLOGIES: MEDIA AND DEVICES



Output technologies differ in their:

- Speed
- Cost
- Portability
- Flexibility
- Storage and retrieval possibilities

■ A COMPARISON OF OUTPUT

ME

Output Method	Advantages	Disadvantages
Printer	• Affordable for most organizations • Flexible in types of output, location, and capabilities • Handles large volumes of output • Highly reliable with little down time	• Still requires some operator intervention • Compatibility problems with computer software • May require special, expensive supplies • Depending on model, may be slow • Environmentally unfriendly
Display screen	• Interactive • Online, real-time transmission • Quiet • Takes advantage of computer capabilities for movement within databases and files • Good for frequently accessed, ephemeral messages	• May require cabling and setup space • Requires system for taking “snapshots” of screen and storing them for future use
Audio output and podcasts	• Good for individual user • Good for transient messages • Good where worker needs hands free • Good if output needs to be widely distributed	• Needs earbuds where output will interfere with other tasks • Has limited application
Mobile devices	• Highly portable • Very interactive using gestures • Zoom is possible	• Screen may be too small for text • Icons and buttons may be confusing • May be lost more easily
Electronic output (email, Web sites, blogs, and RSS feeds)	• Reduces paper • Can be updated very easily • Can be “broadcast” • Can be made interactive	• Is not conducive to formatting (email) • Is difficult to convey context of messages (email) • Web sites need diligent maintenance

Kendall eBook
 Part 4 • The Essentials of Design
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OUTPUT CLASSIFICATIONS

INTERNAL OUTPUT

- used within the corporation
- **example:** all types of report for manager.



NICHOLVILLE TELEPHONE COMPANY, INC.
 P.O. Box 122, Nicholville, New York 12965-0122
 (315) 328-4411

PAGE 1 OF 6

JOE SMITH
 104 WINDY HILL RD.
 NICHOLVILLE NY 12965-0104

AMOUNT DUE \$54.93
 AMOUNT PAID \$

PLEASE SEND PAYMENT TO:
 NICHOLVILLE TELEPHONE COMPANY, INC.
 P.O. Box 122
 Nicholville, New York 12965-0122

TELEPHONE NUMBER: 315-328-5555
 ACCOUNT NUMBER: 1000

SMITH JOE K

PAYMENTS RECEIVED AFTER FEB 28 ARE NOT INCLUDED

PREVIOUS BALANCE	PREVIOUS PAYMENTS	BALANCE FORWARD	CURRENT CHARGES	AMOUNT DUE
\$53.04	\$53.04	\$0.00	\$54.93	\$54.93

CUSTOMER DETAIL

NICHOLVILLE TELEPHONE COMPANY CURRENT CHARGES

CHARGE	AMOUNT
MONTHLY SERVICE FROM MAR 1 THRU MAR 31	31.30
OTHER CHARGES/CREDITS	1.26
AT&T 00288	20.56
VERIZON 00698	1.81
TOTAL CURRENT CHARGES	54.93
TOTAL AMOUNT DUE - PLEASE PAY THIS AMOUNT	\$54.93

EXTERNAL OUTPUT

- used outside the organization
- **example:** pay-checks, annual reports, utility bills, advertisements
- should differs from internal output in its design and appearance.

November 7, 2002 Course Summary Report Pg 1 Of 28					
Fall Semester 2002					
CourseID	Course Name	Units	Total Sects	Total Enroll	Avg. Enroll /Sect
Act102	Accounting Prin.	3	4	300	75
....		...	...	...	...
Bio101	Into to Biology	3	6	600	100
...	...	...	...	...	...
Chm109	Organic Chem	3	2	90	45
...	...	...	...	...	
Mkt114	Prin. Of Marketing	3	2	110	55

TURNAROUND DOCUMENT

- a document that is sent out and then returned;
- some external output is designed to serve double duty as a turnaround document
- **example:** utility bills, warranty cards, etc

MARS
 BIO-MED PROCESSES INC

TECH.:	
CONTACT #:	
BRANCH:	



WARRANTY REGISTRATION CARD

Please complete and return all the information below to info@marsbiomed.com or fax to 905-723-9610 to activate your *LibertyBOSS* Warranty.

INSTALL DATE:	SERIAL #:	OLD:	NEW:
COMPANY:	CONTACT:		
ADDRESS:	TELEPHONE #:		
CITY/STATE:	FAX #:		
ZIP CODE:	EMAIL ADDRESS:		
PRACTICE: (Check one)	PART TIME	AVERAGE	BUSY
# OF DOCTORS	FULL TIME	PART TIME	
TOTAL # OF CHAIRS:		# OF OPS:	# OF HYGIENE ONLY:
# OF HYGIENISTS:		FULL TIME	PART TIME

Follow us on Facebook and check out our new website at www.marsbiomed.com

HOURS OF OPERATION (Times):	MONDAY	TUESDAY	WEDNESDAY	THURSDAY	FRIDAY	SATURDAY	SUNDAY
CLOSED AT LUNCH:	YES	NO	TIME:	WEBSITE:			

www.marsbiomed.com 1-866-594-3648



■ DESIGNING PRINTED OUTPUT REPORT

- Report Design Conventions : WYSIWYG
- Paper Quality, Type and Size
- Organizations strive to reduce the flow of paper and printed reports
 - Users find it handy to view screen output, then print the information they need
 - Printed output is used in **turnaround documents**
- Well organized and consider report design features
 - Functional attributes
 - The heading, page number, date, column heading, grouping data, control breaks)

Example of Good report design

Employee Hours Week ending date: 6/28/2016						report header
Store Number	Employee Name	Position	Regular Hours	Overtime Hours	Total Hours	page header
8	Andres, Marguerite	Clerk	20.0	0.0	20.0	group footer
8	Bogema, Michelle	Clerk	12.5	0.0	12.5	
8	Davenport, Kim	Asst Mgr	40.0	5.0	45.0	
8	Lemka, Susan	Clerk	32.7	0.0	32.7	
8	Ramirez, Rudy	Manager	40.0	8.5	48.5	
8	Ullery, Ruth	Clerk	20.0	0.0	20.0	
Store 8 totals:			165.2	13.5	178.7	
17	De Martini, Jennifer	Clerk	40.0	8.4	48.4	group footer
17	Haff, Lisa	Manager	40.0	0.0	40.0	
17	Rittenbery, Sandra	Clerk	40.0	11.0	51.0	
17	Wyer, Elizabeth	Clerk	20.0	0.0	20.0	
17	Zeigler, Cecille	Clerk	32.0	0.0	32.0	
Store 17 totals:			172.0	19.4	191.4	
Grand totals:			337.2	32.9	370.1	report footer
Page 1						page footer

FIGURE 8-15 The Employee Hours report is a detailed report with control breaks, subtotals, and grand totals. Notice that a report header identifies the report, a page header contains column headings, a group footer contains subtotals for each store, a report footer contains grand totals, and a page footer identifies the page number.

DESIGNING PRINTED OUTPUT

- Types Printed output Reports :
 - **Detailed reports**
 - Print a report line for every record on the master file.
 - Can be quite lengthy
 - **Exception reports**
 - Print a line for all records that match a certain condition.
 - Useful when the user wants specific information
 - **Summary reports**
 - Print one line for a group of records that are used to make decisions.
 - Reports that provide comprehensive data

■ DETAIL REPORT

- Can be quite lengthy, better alternative is to produce an exception report

EMPLOYEE HOURS					PAGE 1
WEEK ENDING DATE: 6/24/05					
STORE NUMBER	POSITION	EMPLOYEE NAME	REGULAR HOURS	OVERTIME HOURS	TOTAL HOURS
8	Asst Mgr	Andres, Marguerite	20.0		20.0
		Bogema, Michelle	12.5		12.5
		Davenport, Kim	40.0	5.0	45.0
	Manager	Lemka, Susan	32.7		32.7
		Ramirez, Rudy	40.0		48.5
		Ullery, Ruth	20.0	8.5	20.0
			165.2	13.5	178.7
STORE 8 TOTALS:					
17	Manager	De Martini, Jennifer	40.0		48.4
		Haff, Lisa	40.0	8.4	40.0
		Rittenbery, Sandra	40.0		51.0
		Wyer, Elizabeth	20.0	11.0	20.0
		Zeigler, Cecille	32.0		32.0
		172.0	19.4	191.4	
STORE 17 TOTALS:					
GRAND TOTALS:		337.2	32.9	370.1	

■ EXCEPTION REPORT

- Useful when the user wants information only on records that might require action.

OVERTIME REPORT WEEK ENDING DATE: 6/24/05			PAGE 1
STORE NUMBER	POSITION	EMPLOYEE NAME	OVERTIME HOURS
8	Asst Mgr Manager	Davenport, Kim	5.0
		Ramirez, Rudy	8.5
	STORE 8 TOTALS:		13.5
11	Manager	Gadzinski, Barbara	10.0
	Clerk	Stites, Carol	12.0
	Asst Mgr	Thompson, Mary Kay	1.5
STORE 11 TOTALS:			23.5
17	Clerk	De Martini, Jennifer	8.4
	Clerk	Rittenbery, Sandra	11.0
STORE 17 TOTALS:			19.4
GRAND TOTAL:			56.4

■ SUMMARY REPORT

- Reports used by individuals at higher levels in the organization include less detail than reports used by lower-level employees

EMPLOYEE HOURS SUMMARY WEEK ENDING DATE: 6/24/05				PAGE 1
STORE NUMBER		REGULAR HOURS	OVERTIME HOURS	TOTAL HOURS
8		181.2	13.5	194.7
11		184.8	23.5	208.3
17		172.0	19.4	191.4
	TOTALS:	538.0	56.4	594.4

■ OUTPUT SCREEN DESIGN

1. Keep the screen simple.
2. Keep the screen presentation consistent.
3. Facilitate user movement among screens.
4. Create an attractive screen.

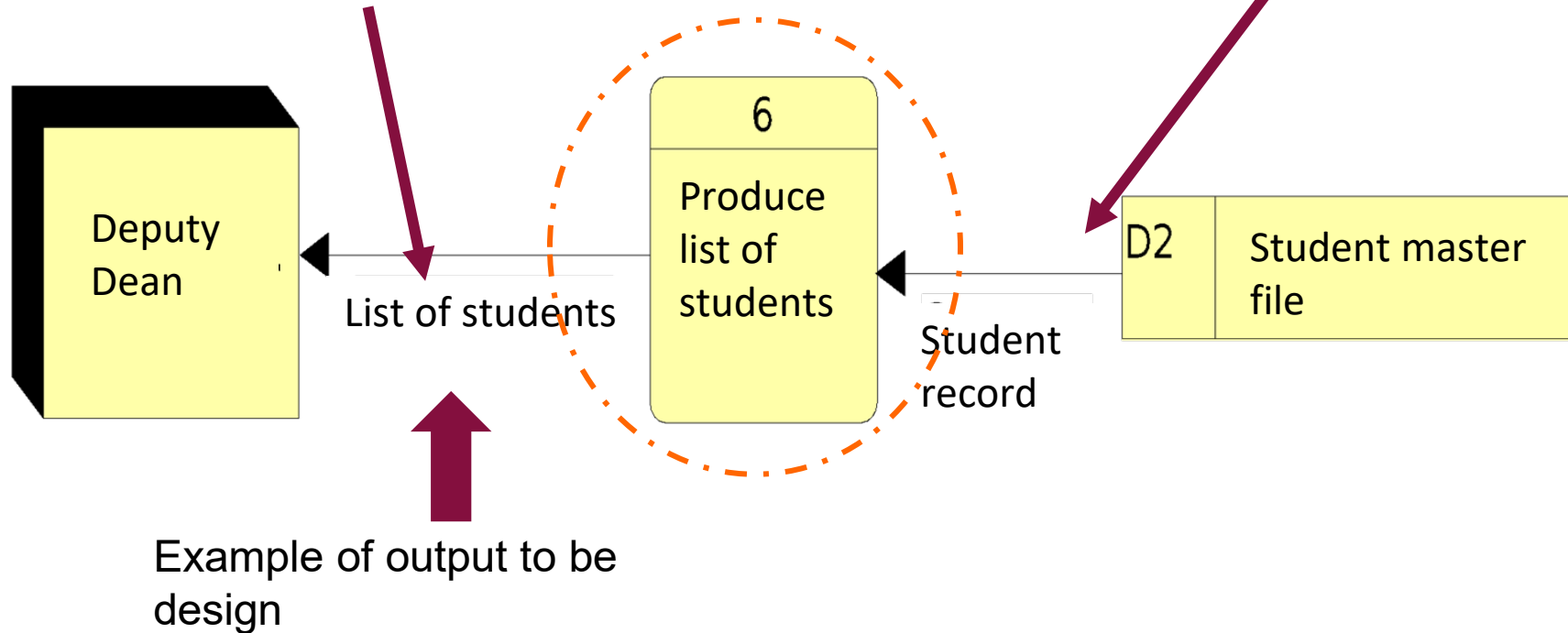


FROM DFD TO OUTPUT DESIGN

Process 6 produce letters to customers:

List of students is a printed

The Student record is a computer record.



FROM DFD TO OUTPUT DESIGN

Example of report from
Process 6

Microsoft Access - [Table1]

File Edit View Tools Window Help

100% Close

Senarai pelajar mengikut major

<i>Major</i>	<i>Nama</i>	<i>Id</i>
<i>GMM</i>	Haida	5
	Limah	4
<i>IS</i>	Ahmad	1
	Liza	2
	Yati	6
<i>KI</i>	Syukor	8
<i>KP</i>	Abu	3

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Ready

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Thank You

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